Adding USGS channel measurements to RiverShapeML

More gage sites, better-constrained targets, and a re-predicted national product with prediction intervals

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Summary

RiverShapeML (Modaresi Rad et al., 2024) learns bankfull width and depth from gage measurements and predicts them for all 2.65 million NHDPlus reaches in CONUS. Its training data come from HYDRoSWOT, which holds acoustic Doppler (ADCP) measurements through 2014. USGS now publishes every channel measurement it makes, by every method (wading, bridge, cableway, boat), through the Water Data OGC API: 4.77 million records. I rebuilt the training targets from those records using the RiverShapeML and AHGestimation conventions, kept the model, features and deployment pipeline unchanged, and asked one question: **does the added data improve the model where it is weakest?**

- **More training sites, mostly where they were scarcest.** Bankfull-width training sites rise from 4,375 to 6,928 (+58%). Sites under 10 km² rise from 66 to 111, and sites of 10–100 km² from 596 to 1,157.
- **The added measurements are sound.** At the same sites, bankfull widths from all measurements and from HYDRoSWOT alone agree at the median (ratio 1.00); depths measured the same way agree within 0.1–2.4%.
- **Held-out error falls in every size class, most in small streams.** Against the same model trained on the existing RiverShapeML data, median absolute error in bankfull width falls from 78% to 41% for sites under 10 km² (bias +67% → +5%), and from 25% to 22% overall; depth improves similarly. The gain comes from the new sites: at existing sites, judged on the original HYDRoSWOT targets, the two models match.
- **A re-predicted national product with 90% prediction intervals.** The retrained model was run through the NOAA-OWP deployment pipeline for every reach, with split-conformal intervals by drainage-area class. It sits closer to the Bieger et al. (2015) regional curves in headwaters and removes most physically implausible values.

The work builds directly on Arash Modaresi Rad's RiverShapeML and the public NOAA-OWP/predict-riverML pipeline, and uses Mike Johnson's AHGestimation package for the target fits. The deployment pipeline is clean and complete enough that its national outputs reproduced exactly from the published models, which made a controlled experiment like this one possible in a few weeks.

1. Motivation

I walked a stretch of Shoal Creek in Austin, Texas, to see how a reach-averaged geometry product compares with the channel on the ground. Over a few hundred meters the channel moves from a walled, rock-lined section with a built low-flow channel, to an incised section held by gabions and a bedrock ledge, to a section rebuilding a small floodplain below a failed grade control. A single bankfull value for the whole 8.2 km reach (COMID 5781223) is the right abstraction for a national product, so that variability is not the subject here.

The gage at the downstream end of the walk is. USGS 08156675 (Shoal Creek at Silverway Drive) has **68 width measurements**. HYDRoSWOT has **2**, both boat measurements from 2009, so the site falls

below the 5-measurement minimum. The other 66, mostly wading measurements, are not in HYDRoSWOT at all. The question this note answers is how common that situation is, and what the model gains when those measurements are used.

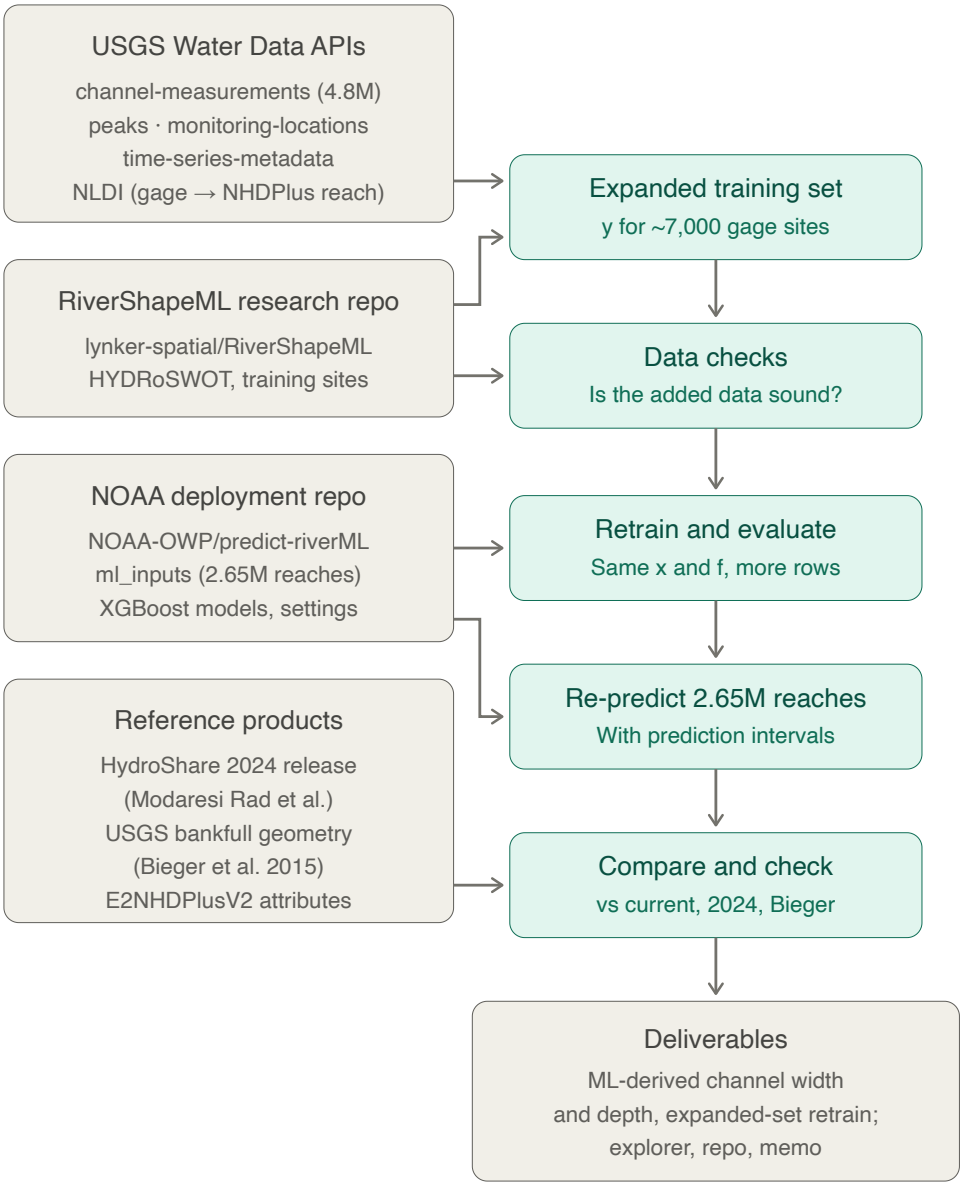


Figure 1. Process. Each step corresponds to scripts in the accompanying repository, and the full table-level lineage (36 tables, 18 operations, each with source, retrieval date and commit) is checked automatically on every run.

2. Data and methods

Measurements. All USGS channel measurements were retrieved from the `channel-measurements` collection of the USGS Water Data OGC API (4,772,063 records), with site metadata and annual peaks from the `monitoring-locations` and `peaks` collections. Gages were linked to NHDPlus reaches with the

NLDI; existing RiverShapeML sites keep their published COMIDs. Candidate new sites are CONUS gages with at least 5 measurements since 2000 and at least 10 years of daily flow (4,501 sites; 3,192 linked to a reach).

Targets. For each site, width, mean depth (area / width) and velocity were converted to SI units and screened with RiverShapeML's sanity rules (width < 4 km, depth < 65 m, Q ≤ 1.1 × maximum annual peak). Hydraulic geometry was then fit with AHGestimation (`mad_filter` → `qva_filter` → `ahg_estimate`), which fits W = aQ^b, D = cQ^f and V = kQ^m jointly under continuity, and each curve was evaluated at the 2-year flood, Q_2. I took Q_2 as the median of the annual peak series, which is the 2-year flood by definition on an annual-maximum basis (England et al., 2019; Institute of Hydrology, 1999); it agrees with RiverShapeML's bankfull flows at the 4,315 shared sites (median ratio 1.02). The target is y = ln(1 + W_bf) with W_bf in meters, as in RiverShapeML, and likewise for depth. Sites where depth decreases with flow (f < 0) were dropped (53 sites).

Target filter. RiverShapeML keeps sites whose width–discharge fit has $R^2 \geq 0.6$. Many well-measured sites fail that test because width barely changes with flow, not because the target is uncertain: 3,636 of the 6,928 width targets here would fail it, with a median standard error of 0.040 on the log target. I replaced it with a precision filter: standard error of the log target at $Q_2 \leq 0.2$, $Q_2 \leq 3 \times$ the largest measured flow (limits extrapolation), and at least 5 measurements.

Model. Features, feature engineering, missing-value handling and inference follow predict-riverML exactly (30 features from `ml_inputs.parquet`). I first reproduced the deployed models' national predictions: on the same reaches, the published output statistics match to six decimals (e.g., minimum and maximum width 1.813823 m and 14,114.583008 m; median 18.43 m). For retraining I kept their XGBoost learning rate and sampling settings but chose tree depth, minimum child weight and number of trees by a regularization sweep with early stopping on a reach-grouped 70/15/15 split (depth 6, min_child_weight 5, about 2,450 trees). This matched the deployed settings' held-out accuracy with roughly a quarter of the trees and much less fit to the training data (test log R^2 0.84 for width, 0.76 for depth). All runs below use the same settings, so only the training data differ.

Experiment. One comparison, with everything but the training data held fixed:

Run	Training sites	Targets built from
A, existing RiverShapeML data	Existing RiverShapeML sites	HYDRoSWOT measurements only
B2, expanded training set	Existing plus new sites	All USGS measurements

Both are scored with 5-fold cross-validation grouped by reach (3 seeds) on the same held-out sites, against the all-measurement targets, which rest on about ten times more measurements per site. Because A never trained on those targets, existing sites are also scored against the HYDRoSWOT-only targets (the same-target check). HYDRoSWOT records maximum rather than mean depth, so A's depth target uses mean depth from the USGS records of the same HYDRoSWOT visits (97% match by date and discharge). Error is reported as ln(predicted / observed), summarized as median absolute error (%), median bias (%), share of sites within a factor of 2, and R^2 in log space.

Prediction intervals. The B2 model was refit on all sites and applied to every reach through the deployment pipeline. Split-conformal 90% prediction intervals (Lei et al., 2018) were computed per drainage-area class from held-out residuals. Coverage was then checked on sites unseen by both the model and the calibration (0.88–0.90 across classes).

3. Results

More sites, concentrated in small drainages

Drainage area (km²)	Published width targets ($R^2 \geq 0.6$)	Expanded set: existing sites	Expanded set: new sites	Expanded total
< 10	66	38	73	111
10–100	596	560	597	1,157
100–1,000	1,766	1,919	876	2,795
1,000–10,000	1,395	1,620	465	2,085
> 10,000	485	577	193	770
Total	**4,375**	**4,719**	**2,209**	**6,928**

Totals include 10 sites with a zero or missing drainage area. Depth targets follow the same pattern (7,017 sites). The gain is largest relative to need: 58% of NHDPlus reaches drain less than 10 km², and those reaches were represented by 66 training sites.

The added measurements are consistent with HYDRoSWOT

Across all target sites, 97% of HYDRoSWOT measurements since 2000 match a record in the USGS collection (within one day, discharge within 3%; 147,074 of 152,100), and in a 300-site sample a typical site has 13 HYDRoSWOT measurements against 142 in total. For width, the curve fit to the added measurements matches the HYDRoSWOT curve over shared flows at the median (+0.6%; 62% of sites within ±10%), and across 3,666 existing sites the bankfull width targets from all measurements and from HYDRoSWOT alone agree at the median (ratio 0.996). For depth, where the same deployment type appears in both, depths agree closely (bridge +0.1%, boat +2.4%, cableway +1.0%). Wading sections are about 33% shallower than bridge sections at the same flows, because they are taken at different cross sections; this is a location difference rather than a measurement error, and the per-site fit pools both. Repeat-measurement scatter sets a noise floor of 8–15% for width and 14–19% for depth. The joint AHG fit changes the median width target by less than 1% relative to a plain log-log fit but corrects poorly conditioned sites (e.g., Walnut Creek: 28.1 m → 19.0 m).

Held-out error falls in every size class, most in small streams

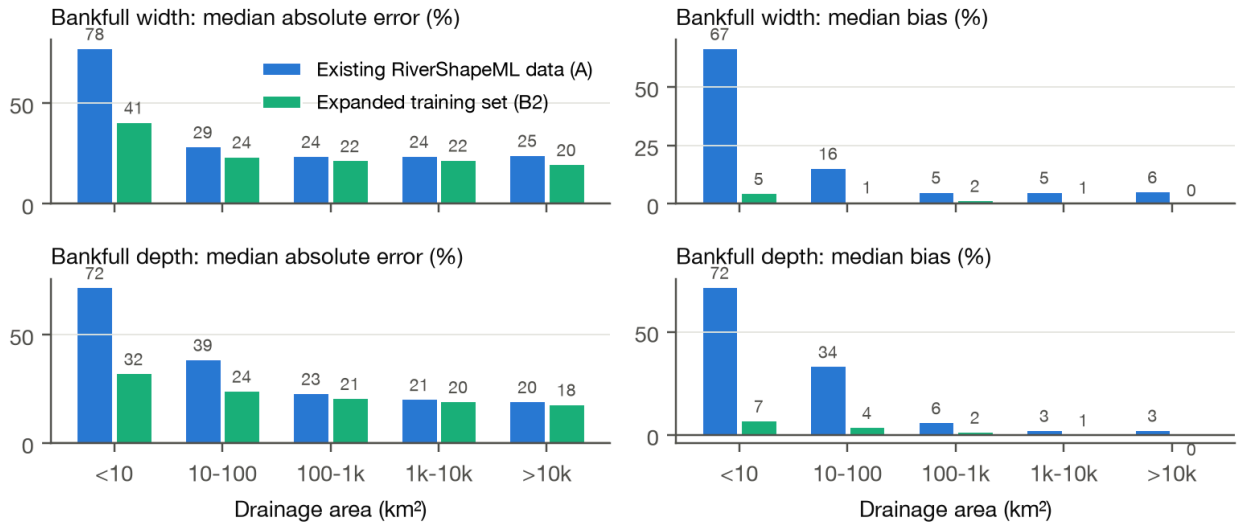


Figure 2. Held-out median absolute error and median bias by drainage-area class: the same model trained on the existing RiverShapeML data (A) and on the expanded training set (B2). Same settings, same held-out sites.

	Width: A → B2	**Depth: A → B2**
Median absolute error, all sites	25.4% → 22.4%	23.8% → 20.9%
Median absolute error, < 10 km²	78% → 41%	72% → 32%
Median bias, < 10 km²	+67% → +5%	+72% → +7%
Within a factor of 2, all sites	89% → 92%	89% → 94%
Log R^2, all sites	0.77 → 0.82	0.66 → 0.76

The model trained on the existing data overpredicts small streams because it has almost no small streams to learn from; the new sites remove most of that bias. Gains are smaller in larger basins.

Where the gain comes from. At sites outside RiverShapeML's training set, the closest proxy here for an unseen reach, median absolute error falls from 32% to 26% for width and from 38% to 25% for depth. At existing sites, judged on the HYDRoSWOT-only targets (the same-target check), the two models match: width 20.4% (A) vs 19.6% (B2), depth 18.2% vs 18.2%. Scored on the all-measurement targets, A also looks worse at existing sites (21.4% vs 19.7% for width), but most of that difference reflects the change in target definition rather than a better model, so I do not count it as a gain. The improvement comes from the new sites, and the expanded model gives up nothing on the original problem. For context, repeat-measurement scatter means no model can do much better than about 10–15% median error on width.

A re-predicted national product

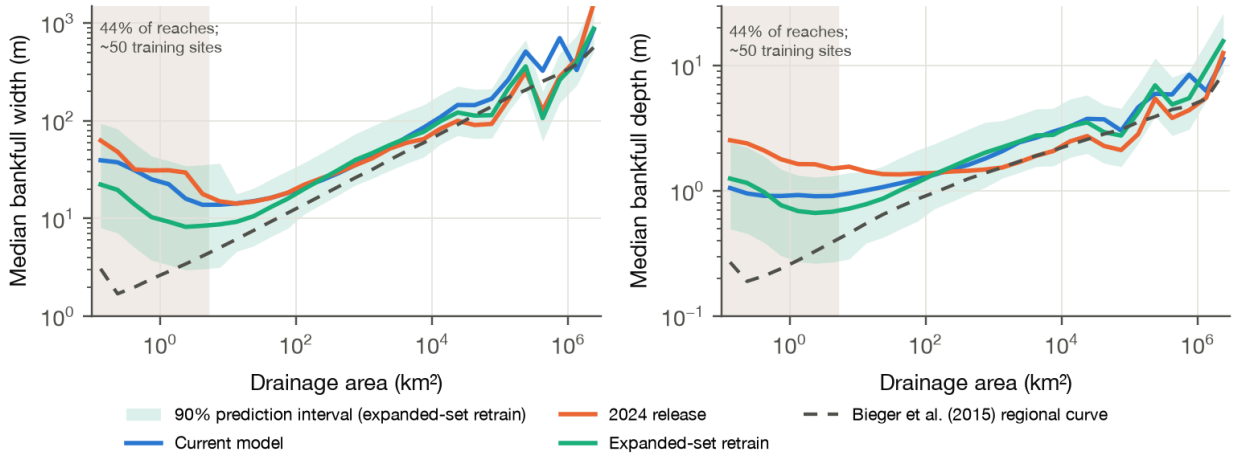


Figure 3. Median bankfull width (left) and depth (right) by drainage area across all CONUS reaches for the current model (predict-riverML), the 2024 HydroShare release, the expanded-set retrain with its 90% prediction interval, and the Bieger et al. (2015) regional curves. The shaded region holds 44% of reaches.

Check (all 2.6 M reaches)	Expanded-set retrain	Current model	2024 release
Width > 1 km	0.002%	0.47%	0.013%
Width / depth > 100	0.09%	1.94%	0.05%
Width halves going downstream	2.2%	3.6%	5.3%
Width–drainage-area exponent (Bieger: 0.36)	0.23	0.15	0.06
Depth–drainage-area exponent (Bieger: 0.25)	0.16	0.12	0.00
Median ratio to Bieger width	1.80	2.88	3.28

Between 100 and 10,000 km² the three ML products agree closely (median widths within about 10%). Above 10,000 km² the current model runs wider (median 152 m, against 122 m for the retrain and 110 m for the 2024 release), and all curves wobble there because few reaches fall in each bin and large rivers of different character (e.g., the Colorado and the Missouri) share them. Below 10 km² they diverge: median widths are 9.1 m (expanded-set retrain), 17.6 m (current model), 26.5 m (2024 release) and 3.5 m (Bieger). Downstream scaling moves toward the regional curves, and implausibly wide reaches nearly disappear.

Depth tells the same story in small streams. Below 10 km² median depths are 0.72 m (expanded-set retrain), 0.93 m (current model), 1.63 m (2024 release) and 0.36 m (Bieger). The 2024 release is nearly flat in depth from the smallest reaches to about 1,000 km² (median 1.5 m), which is the exponent of 0.00 in the table. Between 1,000 and 10,000 km² the retrain and the current model run deeper than both the 2024 release and Bieger (median 2.5 m against 1.7–1.8 m), so the depth improvement is concentrated in small streams. In both panels the Bieger curve rises again below about 0.3 km²; each regional equation is monotone, so this probably reflects which regions those very small reaches fall in. Because most reaches are small, the national median width falls from 18.4 m (current model) to 10.9 m; this is the largest practical change for users of the product. The prediction intervals widen where data are thin: for width they span roughly 0.50–1.9× the prediction above 10 km² and

0.36–4.2× below it. Below 10 km² they rest on about 110 calibration sites, and gaged small streams are not typical of NHDPlus headwaters, so they should be read as indicative there.

The Silverway reach illustrates both the improvement and its limit. The retrain predicts 15.8 m wide and 0.99 m deep (current model 14.8 m / 1.29 m; 2024 release 19.6 m / 1.51 m). The Silverway site itself is not in the expanded training set: its Q_2 (2,700 cfs, 76 m³/s) is 5.2 times the largest measured flow, so its target would be an extrapolation and the filter excludes it.

4. Discussion and limitations

The headwater gap is narrower but not closed. Below about 5 km² there are still only ~50 training sites for 1.16 million reaches. Tree ensembles cannot extrapolate beyond the range of their training targets, so predictions flatten (and in all three products bend upward) at the smallest drainage areas rather than continuing the downstream trend. The ratio to Bieger widths remains 1.8 overall. Part of that gap is definitional: Bieger's field bankfull indicators and a Q_2-based target are not the same quantity, and the Q_2 convention may overstate bankfull flow in small, flashy basins.

Gaged is not ungaged. All accuracy figures are for reaches with gages, which skew toward larger rivers than the network as a whole. The conformal intervals inherit that assumption: they are calibrated on gaged sites and are exchangeable only to the extent that ungaged reaches resemble them.

Some useful data remain unused. About 1,300 candidate gages did not link to a reach through the NLDI, and about 7,000 sites have five or more measurements but less than 10 years of daily flow. The latter could use National Water Model flood frequencies; RiverShapeML's training code already has an NWM flood-frequency option, used there in place of gage floods at the same sites. Most of these sites are small streams (985 under 10 km² and 2,159 at 10–100 km², of those with a known drainage area). In an exploratory check at 7,403 gaged reaches, the NWM 2-year flood ran about 20% below the gage value at the median, and about 50% below under 10 km², where the two barely correlate (squared correlation of logs 0.72 overall, 0.03 for the 137 small sites). That argues for bringing these sites in with lower weight or a flag, and for checking the NWM flows first.

Simplifications to revisit. Three choices here are deliberately simple and worth tightening. (i) Where USGS records a split-flow measurement as one row per channel, the rows are treated as separate points; this affects a minority of sites (about 620 of 6,928 with more than 10% of rows split). (ii) Q_2 uses each gage's full peak record, which often starts decades before the 2000+ measurements, and regulated peaks are kept; both runs share Q_2, so the comparison is unaffected, but absolute target levels are. (iii) Cross-validation holds out whole reaches, not whole basins, so neighbouring gages on the same river can sit on either side of a split; a basin-blocked check would be a useful robustness test.

Two observations I could not resolve from the public record. The current deployed models and the 2024 HydroShare release differ more than I expected: the squared correlation between their log widths is 0.36. Separately, 932 of the 2.65 million reach IDs in the HydroShare file appear more than once (66,341 rows in total). The repeat counts are perfect squares (4, 9, 16, 64, …), which is the signature of a many-to-many merge, and most of these IDs carry two distinct predictions, so they may be two features sharing an ID rather than true copies. For the comparison here each was collapsed to its median; with 932 reaches the effect is negligible. Both points are noted as questions, not findings. Relatedly, the training-target files in the RiverShapeML research repository sit 1.6–1.9 times below the deployed models' predictions at the same sites, while the targets built here match the deployed scale

(ratio 1.0 for depth, 1.13 for width). That suggests the deployed models were trained on a later target set than the one in the repository.

5. Questions and possible next steps

These are offered as starting points for a conversation; the team will know which matter and which have already been considered.

1. **What was the original motivation for training on HYDRoSWOT?** For example, a single, consistent measurement method. The agreement statistics above suggest the other methods can be added with little risk, and I'd be glad to learn about any considerations I haven't seen.
2. **Headwater behavior.** Monotone constraints on drainage area and discharge, or blending with regional curves below a threshold, would keep predictions on a physically sensible trend where data are sparse.
3. **More sites.** Recover the unlinked gages, and use RiverShapeML's NWM flood-frequency option to bring in the ~7,000 short-record sites, mostly small streams.
4. **In-channel geometry.** The same measurements support targets at lower flows (in-channel flow is about 6% of bankfull flow at the median), and these sites are far better sampled there than at Q_2. This is the natural next experiment and maps directly onto the existing in-channel model.
5. **Training settings.** Early stopping and modest regularization reach the same accuracy with about a quarter of the trees, which shortens training and inference.
6. **Release housekeeping.** An updated HydroShare release could carry prediction intervals and resolve the repeated IDs noted above.

I would be glad to prototype any of these within the predict-riverML structure so that the results drop straight into the existing pipeline.

Code and data

Repository: github.com/drmerlot/rivershapeml-more-data. It contains every script, a single `run_all.sh` that rebuilds all results from public sources, a checked data-lineage record, and an interactive notebook (marimo) for exploring sites, targets, runs and the national product. The re-predicted product ("ML-derived channel width and depth, expanded-set retrain", with 90% prediction intervals for every reach) is attached to the repository's v1.0 release as a 38 MB Parquet file, and is regenerated by the pipeline.

Acknowledgments

This work stands on Arash Modaresi Rad's RiverShapeML research and the NOAA Office of Water Prediction's open predict-riverML deployment pipeline, whose published models, inputs and inference code made exact reproduction straightforward. Target fitting uses AHGestimation by Mike Johnson, with contributions from Arash Modaresi Rad. Data are from the USGS (HYDRoSWOT; Water Data APIs; NLDI), the E2NHDPlus attributes, and the Bieger et al. (2015) regional curves (USGS ScienceBase). Any errors of interpretation are mine.

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